FACULTY OF COMPUTING

SEMESTER 2

2023/2024

SECI1143 PROBABILITY AND STATISTICAL DATA ANALYSIS

SECTION 02

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Question 1

(a) Pearson's product-moment correlation coefficient
Spearman's rho rank correlation coefficient

(b)

x	y	x^2	y^2	xy
26	42	676	1764	1092
44	60	1936	3600	2640
53	69	2809	4761	3657
29	47	841	2209	1363
77	91	5929	8281	7007
80	98	6400	9604	7840
20	39	400	1521	780
40	55	1600	3025	2200
67	85	3600	7225	5100
86	104	7396	10816	8949
17	37	289	1369	629
61	77	3721	5929	4697

$\Sigma x = 593$ $\Sigma y = 804$ $\Sigma x^2 = 35597$ $\Sigma y^2 = 60104$ $\Sigma xy = 43969$

$$r = \frac{\Sigma xy - (\Sigma x \Sigma y)/n}{\sqrt{[(\Sigma x^2) - (\Sigma x)^2/n][(\Sigma y^2) - (\Sigma y)^2/n)]}}$$

$$r = \frac{43969 - (593 \times 804)/12}{\sqrt{[(35597) - (593)^2/12][(60104) - (804)^2/12)]}}$$

$= 0.677$

(c) $r = 0.677$ shows that it is relatively moderate positive linear association between number of items and production cost. It can be seen that the production cost increases as the number of items increases.

Question 2

(a)

X	Y	XY	X^2	Y^2
0.27	2	0.540	0.073	4
1.41	3	4.230	1.988	9
2.19	3	6.570	4.796	9
2.83	6	16.980	8.009	36
2.19	4	8.760	4.796	16
1.81	2	3.620	3.276	4
0.85	1	0.850	0.723	1
3.05	5	15.250	9.303	25
$\Sigma X = 14.6$	$\Sigma Y = 26$	$\Sigma XY = 56.8$	$\Sigma X^2 = 32.964$	$\Sigma Y^2 = 104$

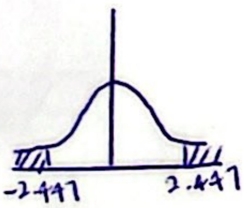
$$r = \frac{(56.8) - [(14.6)(26)]/8}{\sqrt{\left[(32.964) - \frac{(14.6)}{8}\right]\left[(104) - \frac{(26)^2}{8}\right]}} = 0.842$$

(b) $H_0 : \rho = 0$ (no linear correlation)

$H_1 : \rho \neq 0$ (linear correlation exist).

$t_{0.025,6} = \pm 2.447$

$$t = \frac{0.842}{\sqrt{\frac{1-(0.842)^2}{(6-2)}}} = 3.122$$



$\therefore$ ~~t~~ $3.122 > 2.447$, reject Ho. There is sufficient evidence ~~to~~ support that ~~there is a~~ linear correlation exist.

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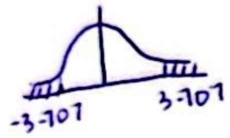
(c) $t_{0.05,6} = 3.707$

$t_{stat} = 3.122$



$\therefore$ fail to reject Ho. there is not sufficient evidence to reject Ho.

QUESTION 3

a] Spearman's Rho Rank Correlation Coefficient: Sentimen Score and Engagement Score.

b] Pearson Correlation Coefficient: Likes and Shares, Comments and Engagement Score.

c)

Engagement Score	Sentiment Score	d	d^2
3	3	0	0
5	4.5	0.5	0.25
1	1.5	-0.5	0.25
6	6	0	0
2	1.5	0.5	0.25
4	4.6	-0.5	0.25
		$\Sigma d = 0$	$\Sigma d^2 = 2$

$$r = 1 - \frac{\cancel{10}6(2)}{10(100^2-1)} = 0.988$$

d) a) likes and comments

x y

$$r = \frac{15180 - (820 \times 100)/6}{\sqrt{[122200 - (820)^2/6][4975 - (165)^2/6]}}$$

$$= \frac{454013}{1531.070795} = 0.988$$

b) likes and share

$$r = \frac{24650 - (820 \times 165)/6}{\sqrt{[122200 - (820)^2/6][4975 - (165)^2/6]}}$$

$$= \frac{2100}{2105.548226} = 0.997$$

∴ The strongest positive correlation among the given pairs is between likes and shares with a Pearson correlation coefficient of 0.997, indicating a very strong linear relationship.

There are no strong negative correlations in provided data.

c) Engagement score and sentiment score

Based on (c), the answer is

$r = 0.988$.

(e)

x	y	x^2	y^2	xy
200	85	40000	7225	17000
100	70	10000	4900	7000
250	90	62500	8100	22500
150	60	22500	3600	9000
220	88	48400	7744	19360
180	75	32400	5625	13500
Σ 1100	Σ 468	Σ 215800	Σ 37194	Σ 88360

$$r = \frac{88360 - 514800/6}{\sqrt{(215800 - 1210000/6)(37194 - 219024/6)}}$$

$$= 0.820$$

$H_0 : \rho = 0$

$H_1 : \rho \neq 0$

$\alpha = 0.05$, d.f $= 4$

$t_{0.025,4} = 2.776$

$$t = \frac{0.820}{\sqrt{\frac{1-0.820^2}{6-2}}} = 2.865$$

$\therefore$ Since $2.776 < 2.865$, reject H_0. There is ~~sufficie~~ sufficient evidence to claim that there is ~~no~~ linear relationship between post length and ~~eg~~ engagement score at the 5% level of significance.

QUESTION 4

a] $\sum x = 95 + 85 + 80 + 70 + 65 = 395$

$\sum y = 85 + 95 + 70 + 75 + 70 = 395$

$\sum xy = 8075 + 8075 + 5600 + 5250 + 4550 = 31\,550$

$\sum x^2 = 9025 + 7225 + 6400 + 4900 + 4225 = 31\,775$

$\therefore$ $\sum x = 395$, $\sum y = 395$, $\sum xy = 31\,550$, $\sum x^2 = 31\,775$

b] $$b_1 = \frac{31\,550 - \frac{(395)(395)}{5}}{31\,775 - \frac{(395)^2}{5}} = \frac{345}{570} = 0.6053$$

$b_0 = 79 - 0.6053(79)$

$= 31.18$

$\therefore$ $b_1 = 0.6053$, $b_0 = 31.18$

c] $\hat{y} = \beta_0 + \beta_1 x$

$\hat{y} = 31.18 + 0.6053x$

d] positive linear relationship

e] given $x = 60$,

$\hat{y} = 31.18 + 0.6053(60) = 67.498$

f] $SSR = (88.6835 - 79)^2 + (82.6305 - 79)^2 + (79.604 - 79)^2 + (73.551 - 79)^2 + (70.5245 - 79)^2$

$= 208.84$

$SST = (85 - 79)^2 + (95 - 79)^2 + (70 - 79)^2 + (75 - 79)^2 + (70 - 79)^2$

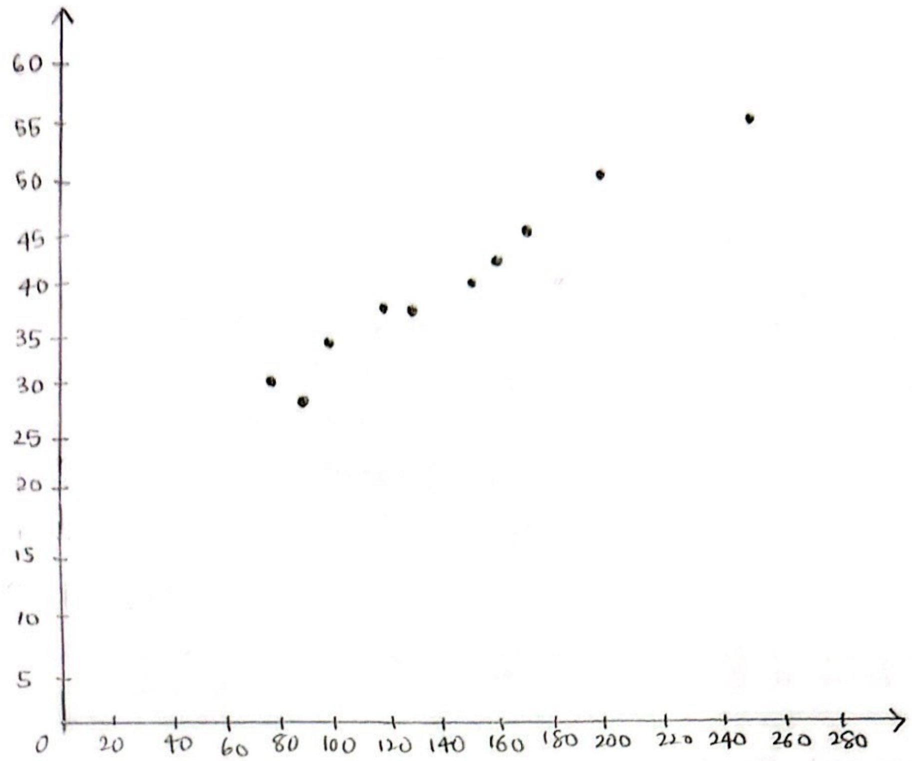
$= 470$

$R^2 = 208.84 \div 470 = 0.4443$

g] the regression equation $\hat{y} = 31.18 + 0.6053x$ moderately fits the data, explaining 44.43 % of the variability in y based on predictor x (with $R^2 = 0.4443$).

Question 5

(a)



A scatter plot and correlation analysis of the data indicates
~~(b)~~ that there is positive relationship between LOC and Development Time.

(b)

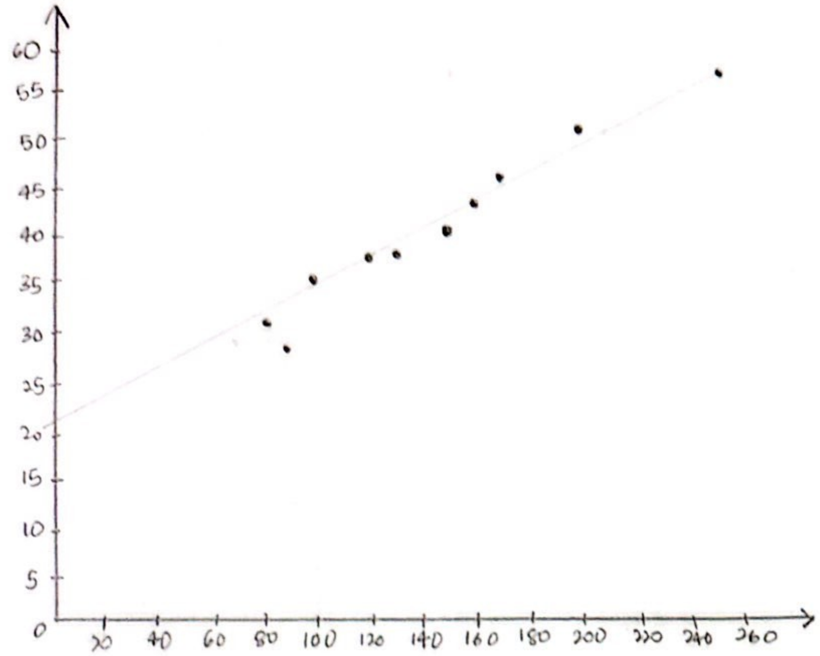
x	y	xy	x^2	y^2
150	40	6000	2250	1600
100	35	3500	10000	1225
200	50	10000	40000	2500
80	30	2400	6400	900
170	45	7650	28900	2025
120	38	4560	14400	1444
160	42	6720	25600	1764
90	28	2520	8100	784
250	55	13750	62500	3025
130	37	4810	16900	1369

$$r = \frac{((61910) - (1450)(400)\,/10}{\sqrt{\left[(235300) - \frac{(1450)^2}{10}\right]\left[(16636) - \frac{(400)^2}{10}\right]}}$$

$r = 0.98$

→ relatively strong positive linear association between x and y.

(c) LOC - independent
DT - dependent



$$b_1 = \frac{(61910) - \frac{(1450)(400)}{10}}{(235300) - \frac{(1450)^2}{10}}$$

$$= \frac{3910}{25050} = 0.156.$$

$$b_0 = 40 - (0.1561)(145)$$
$$= 17.366.$$

$$\hat{y} = 17.366 + 0.156x$$

(d) $R^2 = \frac{SSR}{SST}$ $\frac{(\Sigma \hat{y} - \bar{y})^2}{(y - \bar{y})^2}$

$SST = (40-40)^2 + (35-40)^2 + (50-40)^2 + (30-40)^2 + (45-40)^2 + (38-40)^2 + (42-40)^2 + (28-40)^2 + (55-40)^2 + (37-40)^2$

$= 636$

$SSR = [(17.366 + 0.156(150)) - 40]^2 + [(17.366 + 0.156(100)) - 40]^2 + [(17.366 + 0.156(200)) - 40]^2 + [(17.366 + 0.156(80)) - 40]^2 + [(17.366 + 0.156(170)) - 40]^2 + [(17.366 + 0.156(120)) - 40]^2 + [(17.366 + 0.156(160) - 40]^2 + [(17.366 + 0.156(90)) - 40]^2 + [(17.366 + 0.156(250) - 40]^2 + [(17.366 + 0.156(130)) - 40]^2$

$= 609.619$

$R^2 = \frac{609.619}{636} = 0.959$

An R-squared value of 0.959 means that 95.9% of the variability in the dependent variable (Development Time) can be explained by the independent variabe (LOC).

(e) $\hat{y} = 17.366 i + 0.156 x$

$= 17.36 \quad + 0.156 (180)$

$= 45.446$